A3-PS001-TC171

Android Ransomware-like Activity Early Warning in a Controlled Lab

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Team Introductions

Susnata Maiti
Team Lead

Role
Project coordination and detection engine

Responsible for
Alert rule, response, testing and results

Contribution
Connect the engine to the hook, evaluate events and explain the outcomes

Moupiya Manna
Team Member (Hook)

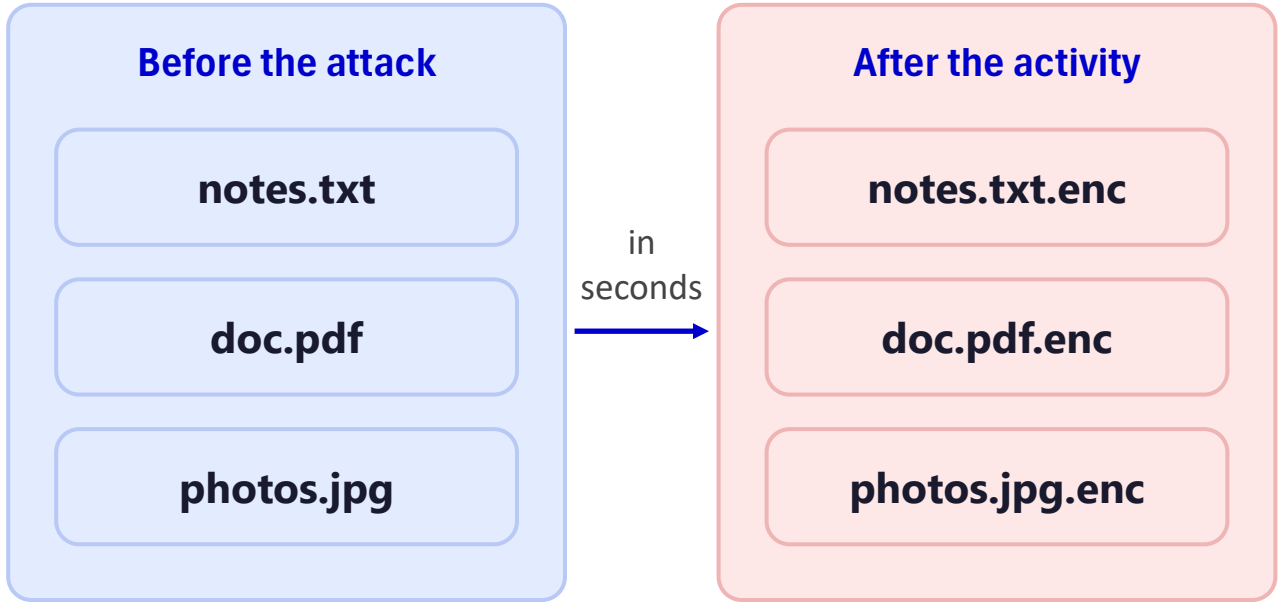
Role
Frida hook and event monitoring

Responsible for
Analysing APK file operations and observing renames

Contribution
Capture file-rename events and send them to the detection engine

Prof. Baisakhi Das presents the simulator workloads in the technical section.

Problem Description



Problem
Zero-day ransomware has no signature, but it renames many files within seconds.

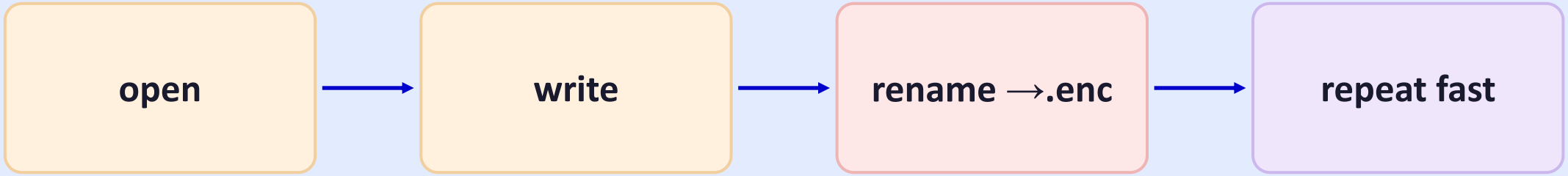
Approach
Hook file renames in a test directory; a rate rule flags bursts.

Outcome
3/3 simulated attacks detected, 0/2 false alarms (lab setting).

Goal: raise an early alert and stop only the test workload

Background

Ransomware file-system pattern



Android sandbox
An app cannot read another app's memory.

⇒

Our observation point
File renames inside one test directory.

Safe by design: emulator only • no real malware • no device-wide monitoring

Literature & RFC Review

Work	Approach	Key signals
Signature antivirus	Match known binaries	Hashes and patterns
HelDroid [3]	Static analysis of mobile ransomware	Lock-screen, encryption and threat-text features
CryptoLock [4]	Dynamic file-activity monitoring	Entropy, file-type change, similarity, deletion
ShieldFS [5]	Filesystem monitoring and recovery	I/O activity models, shadow copies
UNVEIL [6]	Automated dynamic analysis	File I/O traces, screen locking
This work	**Android hook on File.renameTo**	**Rename rate (≥ 10 in 2 s)**

Our focus: a lightweight, explainable rename-rate signal for Android. Entropy and other signals are future work.

No IETF RFC covers app-level file-activity monitoring, so Android security guidance [1] is used.

Proposed Solution

`Red_Team_Simulator.apk`
5 harmless workloads

→

`ueba_hook.js`
Frida hook on File.renameTo

→

`ueba_engine.py`
≥ 10 renames in 2 s ⇒ alert

→

`adb force-stop`
Test app only

Workloads in the APK (test directory)

CryptoLocker-style
simulated attack

DeepTree
simulated attack

PolyWiper
simulated attack

DocAutoSave
benign

PhotoSync
benign

Ethics and safety: no live malware • isolated emulator • only the test directory is touched

Optimization Proposed

Design decisions

Decision	Baseline	Ours
Trigger	Any rename	Renames in the test path
Decision	Single event	≥ 10 events in 2 s
Response	Manual stop	Auto force-stop of test app
Telemetry	None	Delay and files changed

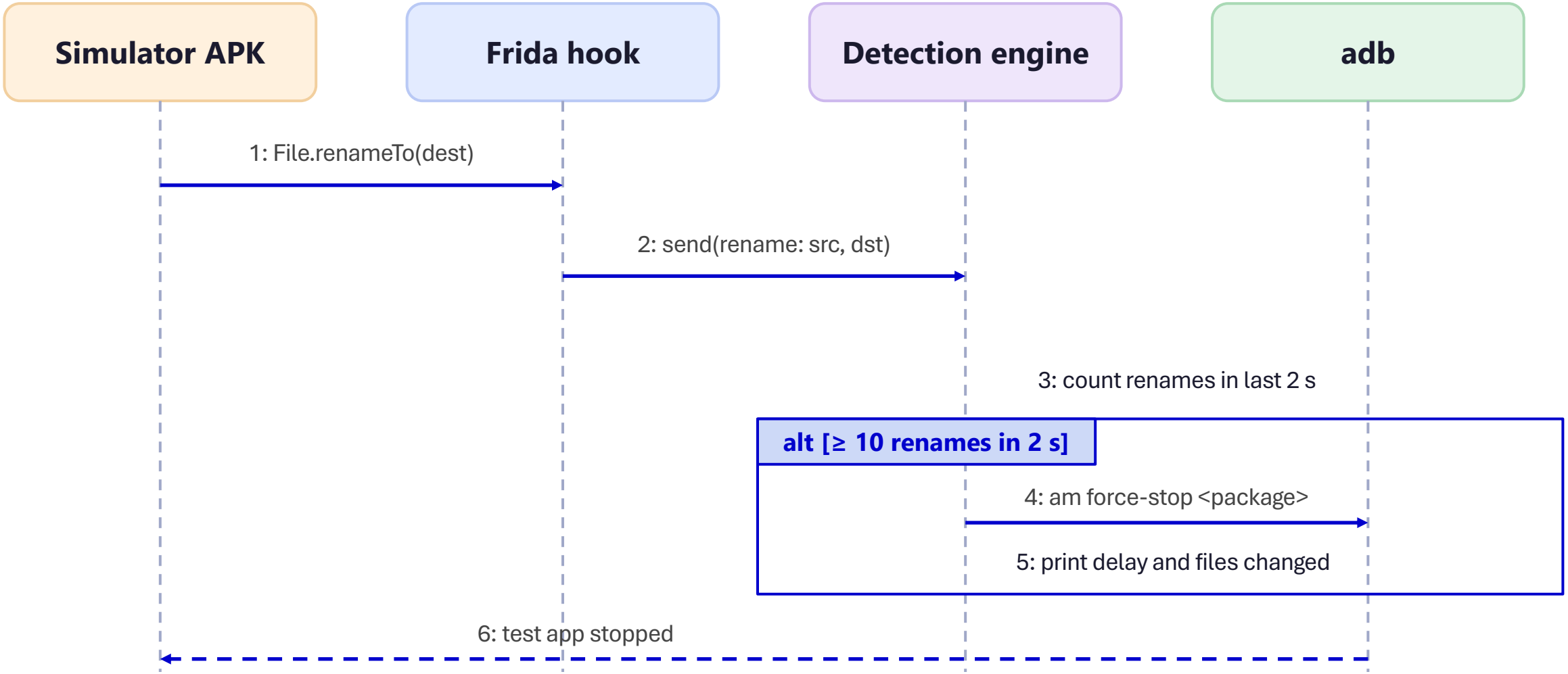
Lab results (5 runs)

Workload	Result	Delay	Files
CryptoLocker-style	Stopped	[s]	[n]
DeepTree	Stopped	[s]	[n]
PolyWiper	Stopped	[s]	[n]
DocAutoSave	Ignored	—	—
PhotoSync	Ignored	—	—

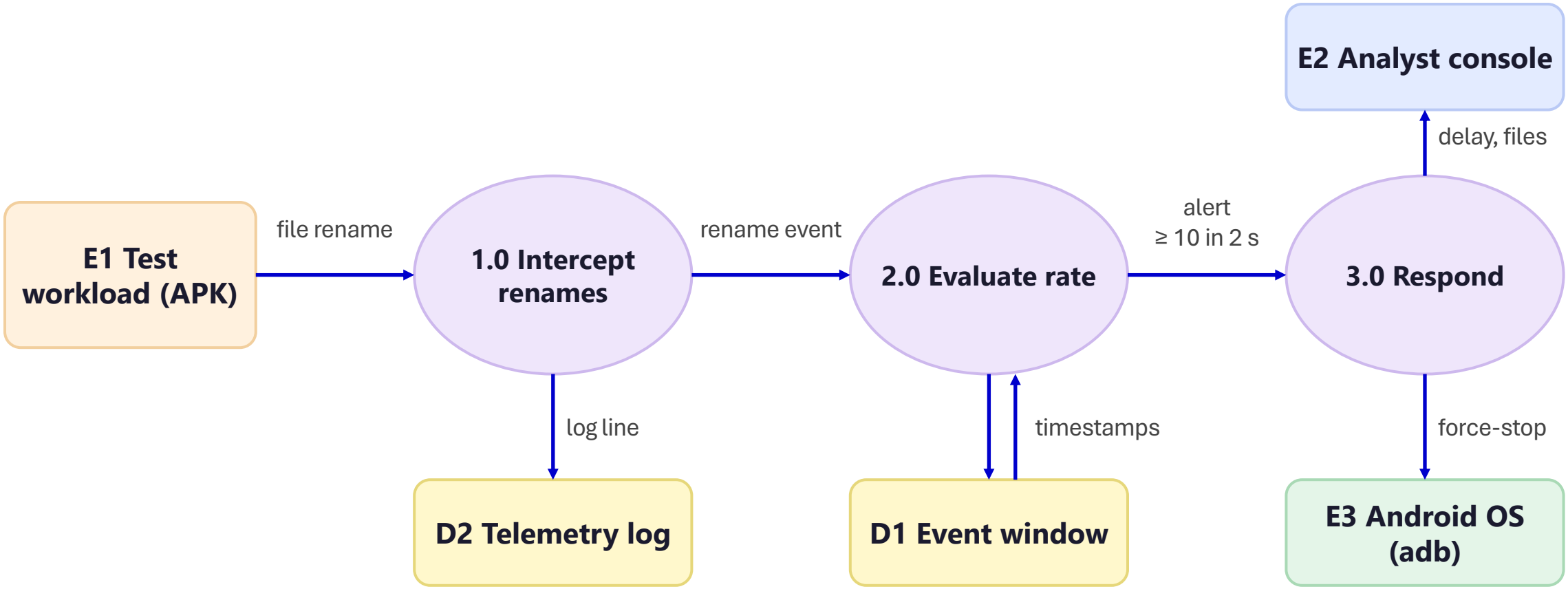
TP 3 • TN 2 • FP 0 • FN 0

Why 10 renames in 2 s? Benign workloads rename at most one file per 2 s, while attacks rename in bursts. 10 sits well above normal use. A threshold sweep is planned.

Solution Architecture & Design



Data Flow Diagram



Rectangle = external entity or data store • Circle = process • Arrow = data flow

Deployment Timeline

Week 1
Lab setup
Emulator, Frida

→

Week 2
Workloads
3 attacks, 2 benign

→

Week 3
Detector
Hook and engine

→

Week 4
Evaluate
Delay, files, repeats

→

Week 5
Demo & docs
Repo, PDF, demo

Known limitations (threat model)

Low and slow
Under 10 renames per 2 s is never flagged.

No rename
Overwrite, or delete-and-write, bypasses the rule.

Native code
libc rename() skips the Java File.renameTo hook.

Lab setup only
Frida and a host PC. Real use needs on-device monitoring.

Future work: libc-level hooks • file-entropy check • multi-signal score • on-device monitor

Bibliography

[1] Android Developers, "Security best practices," developer.android.com/privacy-and-security/security-best-practices (accessed Oct. 2026).

[2] Frida, "Frida: Dynamic instrumentation toolkit," frida.re/docs (accessed Oct. 2026).

[3] N. Andronio, S. Zanero, and F. Maggi, "HelDroid: Dissecting and detecting mobile ransomware," in Proc. RAID, 2015.

[4] N. Scaife, H. Carter, P. Traynor, and K. R. B. Butler, "CryptoLock (and drop it): Stopping ransomware attacks on user data," in Proc. IEEE ICDCS, 2016.

[5] A. Continella et al., "ShieldFS: A self-healing, ransomware-aware filesystem," in Proc. ACSAC, 2016.

[6] A. Kharraz et al., "UNVEIL: A large-scale, automated approach to detecting ransomware," in Proc. USENIX Security, 2016.



Thank You for Your Attention!

Susnata Maiti • Moupiya Manna • Prof. Baisakhi Das